



## Birla Institute of Technology & Science, Pilani

Hyderabad Campus

**FIRST SEMESTER 2026-2027**

### **Course Handout Part II**

**Date: 01.08.2026**

In addition to Part-I (General Handout for all courses appended to the timetable) this portion gives further specific details regarding the course.

**Course No.** : **BITS F113**

**Course Title** : **General Mathematics-1**

**Instructor-in-charge** : **ASHISH TIWARI** (6067-H, ashish.tiwari@pilani.bits-pilani.ac.in)

**Instructors** : **Rakhee** (6067-R, rakhee@pilani.bits-pilani.ac.in)

**1. Course description:** Review of coordinate geometry, Rational Fractions, Complex Numbers and quadratic equations, Matrices and determinants, Progression and series, permutations and combinations, Binomial theorem, Functions: Polynomial, Trigonometric, exponential, logarithmic etc., One variable Calculus: Limit and continuity, Differentiation, Integration. Applications of derivatives and definite integration. Program-specific applications of the above mathematical concepts.

#### **2. Scopes and objective of the course:**

This Course deals with intermediate mathematics that is essential for PCB students who joined BE (E&S engineering). Course covers set theory, Functions, Coordinate geometry, basic algebra & theory of equations, permutations and combinations, Binomial theorem, Trigonometry, One Dimensional Calculus: Limit and continuity, Differentiation, Integration, Application of derivatives and definite integration.

#### **3. Course learning outcomes:**

**CLO1.** Apply concepts of algebra, trigonometry, matrices, and discrete mathematics to solve mathematical problems.

**CLO2.** Analyze and solve problems in coordinate geometry, three-dimensional geometry, and differential calculus.

**CLO3.** Apply integral calculus and related mathematical techniques to solve real-world problems.

#### **4. Text Books and References:**

**TB1.** Mathematics for Class XI: Text book for CBSE national council of educational research and training.

**TB2.** Mathematics for Class XII part I: Text book for CBSE national council of educational research and training.

**TB3.** Mathematics for Class XII part II: Text book for CBSE national council of educational research and training

**RB1.** Thomas Finney: Calculus & analytic geometry 14<sup>th</sup> edition Pearson

**RB2.** Stewart: Calculus early transcendentals 5e 2003 Thomson.



## 5. Lecture Plan:

| Lect No. | Module                        | Lecture Plan                                                                                                                                                                                           | Article                                            | Learning objectives                                                                                                                                                                                                                                                               |
|----------|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1        | Theory of sets                | L1-Sets, operation on sets, finite and infinite set, power set<br>L2- Cartesian product, relations and functions                                                                                       | Chapter 1 & 2 of text book I                       | Students will understand the concept of set theory, basics of functions (domain, codomain, range) and relations                                                                                                                                                                   |
| 2        | Trigonometry                  | L3: Trigonometric Functions and their Identities,<br>L4: Simple Trigonometric Equations,<br>L5: Trigonometric Functions of Sums and Differences of Two Angles,<br>L6: Inverse Trigonometric Functions. | Chapter 3 of text book I,<br>Chapter 2 text book 2 | Students will get exposure to the concepts of right triangle trigonometry, understand the definitions of the inverse trigonometric functions. They will be able to know and apply identities involving the trigonometric functions, Finding solutions of a trigonometric equation |
| 3        | Complex numbers               | L7: Complex numbers,<br>L8: Quadratic equations                                                                                                                                                        | Chapter 4 of text book I                           | Students will get an exposure to the complex number, sum, difference, product and quotient of two complex numbers, conjugate of a complex number, modulus and argument of a complex number, solving quadratic equation.                                                           |
| 4        | Permutations and combinations | L9: Permutations,<br>L10: Combinations                                                                                                                                                                 | Chapter 6 of text book 1                           | Understanding of fundamental principle of counting, deriving relation between permutation and combination.                                                                                                                                                                        |
| 5        | Binomial Theorem              | L11-L12: Binomial theorem for positive integer power                                                                                                                                                   | Chapter 7 of text book 1                           | Exposure to proof of Binomial theorem for positive integral values, finding the value of a given number using binomial theorem, finding general and middle terms.                                                                                                                 |
| 6        | Series                        | L13: Arithmetic progression, geometric progression, Arithmetic mean, geometric mean,<br>L14: Infinite series, infinite geometric series, exponential and logarithmic                                   | Chapter 8 & appendix 1 of text book 1              | Students will learn the concept of sequence and series, progression and their types, definitions and properties of progression, discussion on various types of series.                                                                                                            |



|    |                            |                                                                                                                                                                                                                                                                  |                                                   |                                                                                                                                                                                                                                                           |
|----|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    |                            | series                                                                                                                                                                                                                                                           |                                                   |                                                                                                                                                                                                                                                           |
| 7  | Two dimensional geometry   | L15: Condition for parallelism and perpendicularity of two lines,<br>L16: Angle between lines<br>Equations of line in various forms(slope, intercept, through given two points, slope point, general)<br>L17: distance of a point from a line                    | Chapter 9 of text book 1                          | Understand the concept of parallel and perpendicular lines, linear equations in various forms, solutions of linear equations, geometric interpretations of equations.                                                                                     |
| 8  | Conics                     | L18: Conic sections, eccentricity, latus rectum, Locus,<br>L19: Circle, parabola,<br>L20: Hyperbola, ellipse, pair of lines.                                                                                                                                     | Chapter 10 of text book 1                         | Understanding how circle, ellipse, parabola and hyperbola form the sections of a cone, finding standard equation of circle, parabolas, ellipses and hyperbolas and properties of each of them.                                                            |
| 9  | Three Dimensional Geometry | L21-L22: Three dimensional geometry (distance, equations of line and plane in space, distance of a point from plane)                                                                                                                                             | Chap. 11 of text book 1 & Chap. 11 of text book 3 | Exposure to concept of coordinate axes and coordinate planes in three dimensional spaces, finding coordinates of a point in space, compute the distance between two points, section formula and midpoint of a line segment.                               |
| 10 | Matrices                   | L23-L24: Matrices and determinants; basic properties                                                                                                                                                                                                             | Chapter 3 of Textbook 2.                          | Students will understand the basic matrix theory.                                                                                                                                                                                                         |
| 11 | Differential calculus      | L25-L28: Limits, continuity, differentiability, higher order derivatives,<br>L29-L31: Chain rule<br>Logarithmic differentiation, mean value theorem, Rolle's theorem,<br>L32-L34: Slope of tangents, maxima and minima, indeterminate forms.<br>Program specific | Chap 12 textbook 1.<br>Chap. 5 & 6 text book 2    | Students will be able to learn meaning and concept of limits and continuity, understanding differentiation of a function, various theorems based on limit, continuity and differentiability, applications in various forms, concept of maxima and minima. |



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|----|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    |                   | applications                                                                                                                                                                                                                                           |                              |                                                                                                                                                                         |
| 12 | Integral Calculus | L35-L37: Anti-derivatives and indefinite integrals, Methods of substitution, parts, partial fractions, trigonometric reduction formulas, L38-L40: Fundamental theorem of calculus, Definite integrals, area under curve. Program specific applications | Chapter 7 & 8 of text book 3 | Understanding of integration, methods of integrations, understanding fundamental theorem of calculus, Definite integral and its usage in finding area under the curves. |

6. Evaluation Scheme:

| EC No. | Evaluation Component            | Duration | Weightage (%) | Date & Time     | Nature of Component   |
|--------|---------------------------------|----------|---------------|-----------------|-----------------------|
| 1.     | Quizzes/Class Performance Tests | 20 min   | 20            | To be Announced | Closed Book           |
| 2.     | Mid semester Exam               | 90 min   | 35            | 06/10 AN2       | Closed Book           |
| 4.     | Comprehensive Exam              | 180 min  | 45            | 07/12 AN        | Closed Book/Open Book |

**6. Announcements:** All announcements in relation to the above course will be put up on course page in LMS.

**7. Attendance Policy:** Each student is expected to attend all classes and participate in all evaluation components.

**8. Make up policy:** Make up for the mid-semester/comprehensive examination will be given to genuine cases.

**8. Chamber consultation hours:** TBA

**9. Academic Honesty and Integrity Policy:** Academic honesty and integrity are to be maintained by all the students throughout the semester and no type of academic dishonesty is acceptable.

**10. Criterion for NC:** Below the 10% marks or 30% of the median whichever is lower.

**Instructor In-Charge**

**MATH U110**

